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MATHEMATICAL ANALYSIS / ACHIEVEMENT SETS / FRACTAL GEOMETRY

AN ACHIEVABLE CANTORVAL WITH FULL-DIMENSIONAL BOUNDARY

A variable-base construction and a candidate affirmative solution to Problem 23(i)

$\dim_{\mathrm{H}}(\partial E) = 1$

THE nTH BLOCK

$$B_n = \frac{1}{2^{n-1}(n+2)!}(2n+3, \underbrace{2, \dots, 2}_{n+1})$$

MIXED-RADIX MECHANISM

$$b_n = 2n+4, \qquad |D_n| = b_n, \qquad D_n \pmod{b_n} = \{0, 1, \dots, b_n - 1\}$$

FORMULA-DERIVED LEVEL-1 THROUGH LEVEL-7 CYLINDER GEOMETRY OF A FULL-DIMENSIONAL BOUNDARY SUBSET

A GENUINE GAP AT EVERY BLOCK SCALE

$(R_n, 2/Q_n) \subset \mathbb{R} \setminus E$

Metriq PRISM Laboratory
A research laboratory of Metriq Foundation, Inc.

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An Achievable Cantorval with Full-Dimensional Boundary

A variable-base construction and a candidate affirmative solution to Problem 23(i)

Metriq PRISM Laboratory
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Research status

This manuscript presents a candidate resolution of a published open problem. It has not undergone independent peer review. A December 2025 survey lists as Problem 23(i) the existence of an achievable Cantorval whose boundary has Hausdorff dimension one [2]. A March 2026 paper again describes the integer-dimensional boundary question as open [4]. A targeted literature search completed on 16 July 2026 did not identify a later resolution or the construction below; that search does not establish publication priority. Independent specialist verification is required before the result should be treated as established.

Abstract

For a summable positive sequence, its achievement set is the compact set of all subsums. Głąb and Prus-Wiśniowski ask whether an achievable Cantorval can have boundary of Hausdorff dimension one. We give a variable-base construction intended to answer this question affirmatively. Put

$$Q_0 = 1, \qquad Q_n = \prod_{k=1}^{n}(2k+4) = 2^{n-1}(n+2)!,$$

and concatenate the finite blocks

$$B_n = \left(\frac{2n+3}{Q_n}, \underbrace{\frac{2}{Q_n}, \dots, \frac{2}{Q_n}}_{n+1 \text{ copies}} \right), \qquad n \geq 1.$$

The block subset sums form a complete residue system modulo the mixed radix $b_n = 2n+4$. This implies that the resulting achievement set E has nonempty interior. A sharp tail estimate produces a distinct gap after every block, so the Guthrie–Nymann classification implies that E is a Cantorval. We then construct a separated Moran-type subset $C \subseteq \partial E$ with $(n+1)!$ level-n cylinders and scale Q_n^{-1}. A direct mass-distribution argument proves $\dim_{\mathrm{H}} C = 1$, and therefore $\dim_{\mathrm{H}}(\partial E) = 1$. The proof is analytic; the accompanying code performs finite exact-arithmetic consistency checks but is not used as evidence for the infinite claims.

2020 Mathematics Subject Classification. Primary 40A05; Secondary 28A80, 11B05, 28A78.
Keywords. Achievement set; set of subsums; Cantorval; Hausdorff dimension; boundary; mixed-radix expansion; Moran set.

Candidate main result. There exists a summable positive sequence whose achievement set E is a Cantorval and satisfies

$$\dim_{\mathrm{H}}(\partial E) = 1.$$

The explicit sequence is the concatenation of the blocks B_n displayed above.

1 Introduction

Let $x = (x_k)_{k\geq 1}$ be a summable sequence of positive real numbers. Its *achievement set*, or *set of subsums*, is

$$\mathscr{E}(x) = \left\{ \sum_{k=1}^{\infty} \varepsilon_k x_k \;:\; \varepsilon_k \in \{0, 1\} \right\}.$$

The set $\mathscr{E}(x)$ is compact and perfect. The classification theorem of Guthrie and Nymann states that it is either a finite union of closed intervals, a Cantor set, or a Cantorval [3]; Nymann and Sáenz later supplied a complete corrected proof [5]. Here a *Cantorval* means the third topological type: a compact set with nonempty interior that is not a finite union of intervals and whose boundary is a Cantor set.

Recent work has begun to quantify the fractal geometry of Cantorval boundaries. Pratsiovytyi and Karvatskyi calculated a fractional Hausdorff dimension for the boundary of the Guthrie–Nymann example and related families [6]. Karvatskyi subsequently obtained the family

$$\dim_{\mathrm{H}}(\partial K_\ell) = \frac{\log(2\ell + 1)}{\log(2\ell + 2)},$$

which approaches one as $\ell \to \infty$ but is strictly less than one for every fixed parameter [4]. The 2025 survey of Głąb and Prus-Wiśniowski records the following question [2, Problem 23]:

Problem 23(i)

Does there exist an achievable Cantorval E such that

$$\dim_{\mathrm{H}}(\mathrm{Fr}\, E) = 1?$$

The construction below replaces a fixed self-similar parameter with a slowly increasing mixed radix. Each block is a finite Guthrie–Nymann-type digit system. Three features are engineered simultaneously:

1. the block digits form a complete residue system, forcing interior;
2. the tail remains shorter than the smallest earlier term, forcing infinitely many gaps;
3. the number of protected boundary branches grows at asymptotically the same logarithmic rate as the inverse geometric scale, forcing boundary dimension one.

2 Preliminaries

Definition 2.1. For $n \geq 1$, let

$$b_n = 2n + 4, \qquad Q_0 = 1, \qquad Q_n = \prod_{k=1}^{n} b_k = 2^{n-1}(n + 2)!.$$

For a sequence of finite digit sets $D_n \subset \mathbb{N}_0$, write

$$K(D_n, Q_n) = \left\{ \sum_{n=1}^{\infty} \frac{d_n}{Q_n} \;:\; d_n \in D_n \right\},$$

whenever the series is uniformly convergent over the allowed digits.

We use the elementary telescoping identity

$$\frac{b_n - 1}{Q_n} = \frac{1}{Q_{n-1}} - \frac{1}{Q_n}, \qquad \sum_{k=n+1}^{\infty} \frac{b_k - 1}{Q_k} = \frac{1}{Q_n}. \tag{1}$$

We also use the following standard mass-distribution principle; see, for example, Falconer [1].

Lemma 2.2 (Mass-distribution principle). *Let $F \subset \mathbb{R}$ be compact and let μ be a probability measure supported on F. If for some $s > 0$ and $C > 0$,*

$$\mu(I) \le C|I|^s$$

for every sufficiently short interval I, then $\dim_{\mathrm{H}} F \ge s$.

3 The block construction

For each $n \ge 1$, define the block

$$B_n = \left(\frac{2n+3}{Q_n}, \underbrace{\frac{2}{Q_n}, \dots, \frac{2}{Q_n}}_{n+1 \text{ copies}} \right). \tag{2}$$

Let x be the infinite sequence obtained by concatenating $B_1, B_2, \dots$, and let

$$E = \mathscr{E}(x).$$

3.1 Basic properties

The block is already ordered nonincreasingly. Moreover,

$$\frac{2}{Q_n} > \frac{2n+5}{Q_{n+1}} = \frac{2n+5}{(2n+6)Q_n},$$

so the concatenated sequence is nonincreasing, apart from the intended repetitions of the small term.

The total of block n is

$$\frac{2n+3+2(n+1)}{Q_n} = \frac{4n+5}{Q_n} = \frac{2b_n - 3}{Q_n}.$$

Consequently, by (1),

$$\sum_{k=1}^{\infty} x_k = \sum_{n=1}^{\infty} \frac{2b_n - 3}{Q_n} < 2 \sum_{n=1}^{\infty} \frac{b_n - 1}{Q_n} = 2. \tag{3}$$

Thus x is a summable positive sequence.

3.2 The block digit sets

Within block n, selecting j of the $n+1$ equal small terms contributes $2j/Q_n$. Selecting the large term as well contributes $(2n+3+2j)/Q_n$. Hence the integer digit set of all block subsums is

$$D_n = \{0, 2, 4, \dots, 2n+2\} \cup \{2n+3, 2n+5, \dots, 4n+5\}. \tag{4}$$

It follows that

$$E = \left\{ \sum_{n=1}^{\infty} \frac{d_n}{Q_n} \, : \, d_n \in D_n \right\}. \tag{5}$$

Lemma 3.1 (Complete residue property). *For every $n \ge 1$, D_n has exactly b_n elements and contains one representative of each residue class modulo b_n.*

Table 1: The first four blocks and their mixed-radix data. Numerators are displayed before division by Q_n.

n	b_n	Q_n	block numerators	D_n
1	6	6	$(5, 2, 2)$	$\{0, 2, 4, 5, 7, 9\}$
2	8	48	$(7, 2, 2, 2)$	$\{0, 2, 4, 6, 7, 9, 11, 13\}$
3	10	480	$(9, 2, 2, 2, 2)$	$\{0, 2, 4, 6, 8, 9, 11, 13, 15, 17\}$
4	12	5760	$(11, 2, 2, 2, 2, 2)$	$\{0, 2, \dots, 10, 11, 13, \dots, 21\}$

Proof. The first part of (4) contains the $n + 2$ even residues

$$0, 2, \dots, b_n - 2.$$

The second part contains $n + 2$ odd integers. Reduced modulo b_n, they are

$$b_n - 1, 1, 3, \dots, b_n - 3.$$

Together these are all b_n residue classes, each exactly once. □

4 The achievement set has interior

For $N \geq 1$, define the integer prefix set

$$P_N = \left\{ \sum_{n=1}^{N} d_n \frac{Q_N}{Q_n} \, : \, d_n \in D_n \right\}.$$

Lemma 4.1 (Mixed-radix bijection). *Reduction modulo Q_N maps P_N bijectively onto $\mathbb{Z}/Q_N\mathbb{Z}$.*

Proof. Suppose two digit words satisfy

$$\sum_{n=1}^{N} d_n \frac{Q_N}{Q_n} \equiv \sum_{n=1}^{N} d_n' \frac{Q_N}{Q_n} \pmod{Q_N}.$$

Reducing modulo b_N leaves $d_N \equiv d_N' \pmod{b_N}$, because every earlier coefficient Q_N/Q_n is divisible by b_N. By theorem 3.1, D_N has a unique representative of each residue, so $d_N = d_N'$. Subtract the equal last digits and divide the congruence by b_N. Repeating the argument gives $d_n = d_n'$ for every $n \leq N$.

Thus the residue map is injective on the set of digit words. The number of such words is

$$\prod_{n=1}^{N} |D_n| = \prod_{n=1}^{N} b_n = Q_N,$$

which equals the number of residue classes modulo Q_N. Hence the map is bijective. □

Proposition 4.2. *The achievement set E has nonempty interior.*

Proof. Fix N and $r \in \{0, 1, \dots, Q_N - 1\}$. By theorem 4.1, there is a prefix integer $p \in P_N$ such that

$$p \equiv r \pmod{Q_N}.$$

Write $p = mQ_N + r$. Choosing zero digits after level N shows $p/Q_N \in E$. By (3),

$$0 \leq \frac{p}{Q_N} < 2,$$

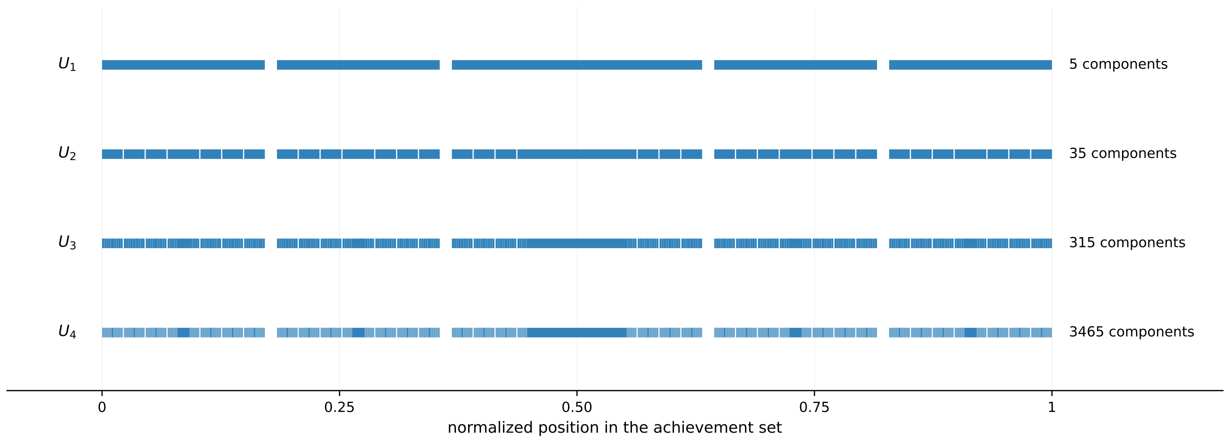

Figure 1: Exact finite-level outer approximations U_N. The increasingly fine components coexist with persistent macroscopic gaps; the central overlap region is consistent with the interior mechanism in theorem 4.2. Positions are normalized by the total sum.

so $m \in \{0, 1\}$. Therefore

$$\frac{r}{Q_N} = \frac{p}{Q_N} - m \in E \cup (E - 1).$$

It follows that every mixed-radix grid

$$\left\{\frac{r}{Q_N} : 0 \le r < Q_N\right\}$$

is contained in $E \cup (E - 1)$. Their union is dense in $[0, 1]$. Since $E \cup (E - 1)$ is closed, it contains $[0, 1]$.

If E had empty interior, then the closed sets E and $E - 1$ would both be nowhere dense. Their finite union could not contain a nondegenerate interval. This contradiction proves $\operatorname{int} E \neq \emptyset$. □

The finite unions

$$U_N = \bigcup_{p \in P_N} \left[\frac{p}{Q_N}, \frac{p}{Q_N} + R_N\right]$$

are exact outer approximations of E, where R_N is the full tail after block N. Figure 1 shows their first four levels.

5 Infinitely many gaps and the Cantorval conclusion

Let

$$R_n = \sum_{k=n+1}^{\infty} \frac{2b_k - 3}{Q_k}$$

be the sum of every term occurring after block n.

Lemma 5.1 (Exact tail identity). *For every $n \ge 0$,*

$$R_n = \frac{2}{Q_n} - \sum_{k=n+1}^{\infty} \frac{1}{Q_k} < \frac{2}{Q_n}. \tag{6}$$

In particular,

$$\frac{2}{Q_n} - R_n = \sum_{k=n+1}^{\infty} \frac{1}{Q_k} > 0. \tag{7}$$

Proof. Because $2b_k - 3 = 2(b_k - 1) - 1$, the telescoping identity (1) gives

$$R_n = 2\sum_{k=n+1}^{\infty} \frac{b_k - 1}{Q_k} - \sum_{k=n+1}^{\infty} \frac{1}{Q_k} = \frac{2}{Q_n} - \sum_{k=n+1}^{\infty} \frac{1}{Q_k}.$$

The final series is strictly positive, proving both assertions. □

Proposition 5.2. *For every $n \geq 1$,*

$$G_n = \left(R_n, \frac{2}{Q_n}\right)$$

is a gap of E. These gaps are pairwise distinct.

Proof. A subsum using no term from the first n blocks is at most R_n. Any subsum using at least one term from those blocks is at least $2/Q_n$, because $2/Q_n$ is the smallest individual term among the first n blocks. Therefore $G_n \cap E = \emptyset$.

Both endpoints belong to E: R_n is obtained by choosing every term after block n and no earlier term, while $2/Q_n$ is obtained by choosing one small term from block n and no other term. Hence G_n is a component of the complement inside the convex hull of E, and its positive length is given exactly by (7).

Moreover, R_n contains the entire $(n+1)$st block, so

$$R_n > \frac{2b_{n+1} - 3}{Q_{n+1}} > \frac{2}{Q_{n+1}}.$$

Thus G_{n+1} lies strictly to the left of G_n, and the gaps are distinct. □

Corollary 5.3. *The achievement set E is a Cantorval.*

Proof. By theorem 4.2, E is not a Cantor set. By theorem 5.2, it has infinitely many gaps, so it is not a finite union of intervals. The Guthrie–Nymann classification theorem, with the corrected proof of Nymann and Sáenz, leaves only the Cantorval alternative [3, 5]. □

6 A protected subset of the boundary

Define the lower-even digit set

$$A_n = \{0, 2, 4, \dots, 2n\} \subset D_n \tag{8}$$

and the compact set

$$C = \left\{ \sum_{n=1}^{\infty} \frac{a_n}{Q_n} \;:\; a_n \in A_n \right\} \subset E. \tag{9}$$

The omission of the final even digit $2n + 2$ leaves a protected next branch at every level.

For a finite word $a_1, \dots, a_m$, put

$$s_m(a) = \sum_{k=1}^{m} \frac{a_k}{Q_k}$$

and define its full achievement-set branch by

$$E[a_1, \dots, a_m] := s_m(a) + \left\{ \sum_{k=m+1}^{\infty} \frac{d_k}{Q_k} \;:\; d_k \in D_k \right\}. \tag{10}$$

Every such branch is contained in the parent interval

$$E[a_1, \dots, a_m] \subseteq [s_m(a), s_m(a) + R_m]. \tag{11}$$

Lemma 6.1 (Isolation of lower-even cylinders). *Let $a_k \in A_k$ for $1 \le k \le m$. The interval*

$$I(a_1, \dots, a_m) = [s_m(a), s_m(a) + R_m]$$

meets no branch of E whose first m digits differ from $(a_1, \dots, a_m)$.

Proof. Consider another digit word and let $j \le m$ be the first index at which it differs. Because $a_j \in A_j$, no element of D_j equals $a_j \pm 1$; hence the two jth digits differ by at least 2.

If the other digit is larger, the other branch begins at least $2/Q_j$ above the common prefix plus a_j/Q_j, while every continuation of the a_j branch lies at most R_j above it. By theorem 5.1, $R_j < 2/Q_j$, so the branches are disjoint. The case of a smaller digit is symmetric. □

Proposition 6.2. *The set C is contained in the boundary of E.*

Proof. Fix

$$x = \sum_{k=1}^{\infty} \frac{a_k}{Q_k} \in C, \qquad s_n = \sum_{k=1}^{n} \frac{a_k}{Q_k}.$$

Since $a_n \le 2n$, the next even digit $a_n + 2$ belongs to D_n. Within the isolated parent cylinder determined by $a_1, \dots, a_{n-1}$, the branch with digit a_n lies in

$$[s_n, s_n + R_n],$$

while every larger nth digit begins at or above $s_n + 2/Q_n$. Every smaller nth digit ends below s_n, because $R_n < 2/Q_n$. By theorem 6.1, no branch with a different parent prefix can enter this region. Therefore

$$J_n = \left(s_n + R_n,\ s_n + \frac{2}{Q_n}\right) \tag{12}$$

is a genuine gap of E.

The point x lies in $[s_n, s_n + R_n]$, and $R_n \to 0$. Hence gaps of the complement occur arbitrarily close to x. Since $x \in E$ and E is closed, $x \in \partial E$. □

Figure 2 displays the first six cylinder levels of C. The number of level-n cylinders is

$$M_n = \prod_{k=1}^{n} |A_k| = \prod_{k=1}^{n} (k+1) = (n+1)!.$$

7 The boundary has Hausdorff dimension one

Because every digit in A_n is even, write

$$C = 2F, \qquad F = \left\{ \sum_{n=1}^{\infty} \frac{j_n}{Q_n} \,:\, 0 \le j_n \le n \right\}.$$

Scaling by 2 does not change Hausdorff dimension. We prove $\dim_{\mathrm{H}} F = 1$ directly.

Let

$$M_n = (n+1)!, \qquad \tau_n = \sum_{k=n+1}^{\infty} \frac{k}{Q_k}.$$

The level-n cylinders of F have tail diameter τ_n. By (1),

$$\tau_n < \sum_{k=n+1}^{\infty} \frac{b_k - 1}{Q_k} = \frac{1}{Q_n}. \tag{13}$$

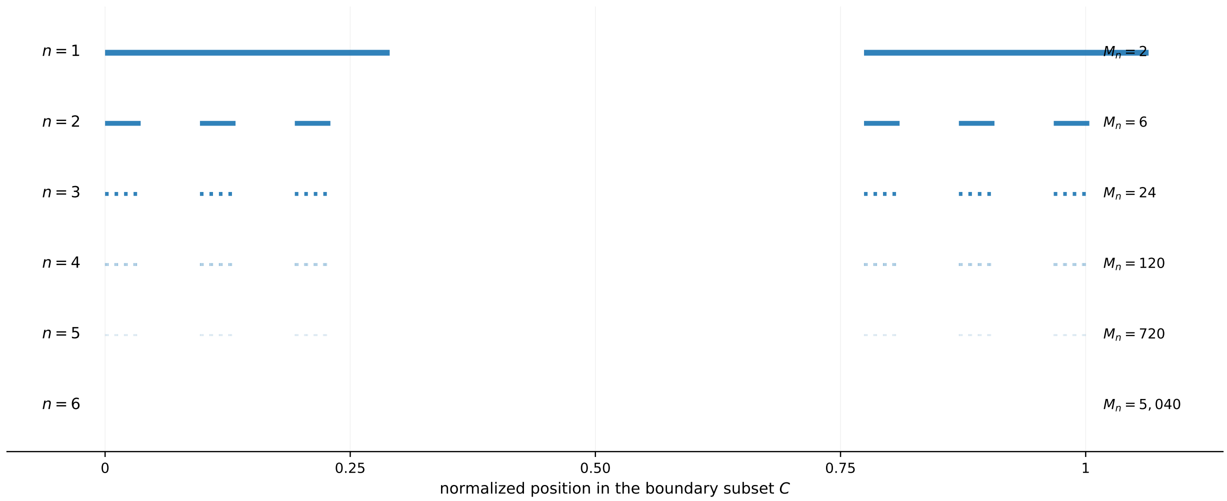

Figure 2: Finite-level cylinders of the protected set $C \subseteq \partial E$. The branch count is $M_n = (n+1)!$, while the geometric denominator is $Q_n = 2^{n-1}(n+2)!$. The drawing uses the exact mixed-radix positions, normalized to the convex hull of C.

Distinct level-n prefix points differ by an integer multiple of $1/Q_n$, so they are separated by at least $1/Q_n$. Thus the level cylinders are pairwise disjoint and separated except possibly at a scale smaller than their common grid spacing.

Give each digit $j_n \in \{0, 1, \dots, n\}$ equal probability, take the corresponding product probability measure on the coding space, and push it forward under the coding map to obtain μ on F. The separation in (13) makes the coding map injective at every finite prefix. Consequently every level-n image cylinder has exactly the product mass

$$\mu(\text{level-}n \text{ cylinder}) = \prod_{k=1}^{n} \frac{1}{k+1} = \frac{1}{M_n}.$$

Proposition 7.1. *For every $0 < s < 1$, there is a constant C_s such that*

$$\mu(I) \le C_s |I|^s$$

for every sufficiently short interval I. Consequently, $\dim_{\mathrm{H}} F = 1$.

Proof. Fix $s \in (0, 1)$ and define

$$\alpha_n = \frac{Q_n^s}{M_n}.$$

Since

$$\frac{\alpha_n}{\alpha_{n-1}} = \frac{b_n^s}{n+1} = \frac{(2n+4)^s}{n+1} \longrightarrow 0,$$

the quantity

$$K_s = \sup_{n \ge 0} \alpha_n$$

is finite, where $\alpha_0 = 1$.

Let I be an interval of length $r < 1$ and choose n so that

$$\frac{1}{Q_n} \le r < \frac{1}{Q_{n-1}}. \tag{14}$$

Put $t = rQ_n$, so $1 \le t < b_n$. By (13), the level-n cylinder starts are $1/Q_n$-separated and each cylinder has diameter less than $1/Q_n$. Therefore I meets at most $t + 2 \le 3t$ level-n cylinders, giving

$$\mu(I) \le \frac{3t}{M_n}. \tag{15}$$

At level $n-1$, the interval I meets at most three cylinders, so

$$\mu(I) \le \frac{3}{M_{n-1}}. \tag{16}$$

There are two cases. If $t \ge n+1$, then

$$r = \frac{t}{b_n Q_{n-1}} \ge \frac{n+1}{b_n Q_{n-1}} \ge \frac{1}{3Q_{n-1}},$$

because $b_n/(n+1) \le 3$. By (16),

$$\mu(I) \le 3\alpha_{n-1} Q_{n-1}^{-s} \le 3^{s+1} K_s r^s.$$

If $1 \le t < n+1$, then (15) gives

$$\mu(I) \le 3\alpha_n t^{1-s} r^s \le 3\alpha_n (n+1)^{1-s} r^s.$$

Using

$$\alpha_n (n+1)^{1-s} = \alpha_{n-1} \left(\frac{b_n}{n+1} \right)^s \le 3^s K_s,$$

we again obtain

$$\mu(I) \le 3^{s+1} K_s r^s.$$

The mass-distribution principle now yields $\dim_{\mathrm{H}} F \ge s$. Since this holds for every $s < 1$ and $F \subset \mathbb{R}$, we conclude $\dim_{\mathrm{H}} F = 1$. □

The logarithmic growth underlying the proof is transparent:

$$M_n = (n+1)!, \qquad Q_n = 2^{n-1}(n+2)!,$$

so

$$\frac{\log M_n}{\log Q_n} = \frac{\log(n+1)!}{\log(n+1)! + (n-1)\log 2 + \log(n+2)} \longrightarrow 1.$$

The measure argument above upgrades this heuristic cylinder ratio to a rigorous lower bound.

Theorem 7.2 (Main theorem). *Let x be the positive summable sequence obtained by concatenating the blocks (2), and let $E = \mathscr{E}(x)$. Then E is a Cantorval and*

$$\boxed{\dim_{\mathrm{H}}(\partial E) = 1.}$$

Proof. By theorem 5.3, E is a Cantorval. By theorem 6.2, $C \subseteq \partial E$. By theorem 7.1,

$$\dim_{\mathrm{H}} C = \dim_{\mathrm{H}} F = 1.$$

Therefore

$$1 = \dim_{\mathrm{H}} C \le \dim_{\mathrm{H}}(\partial E) \le 1,$$

which proves the claim. □

Corollary 7.3. *Subject to independent verification and the literature-status caveat stated above, theorem 7.2 gives an affirmative answer to Problem 23(i) of Głąb and Prus-Wiśniowski [2].*

8 Further observations and limitations

8.1 The full-dimensional witness is null

The protected set C need not be a fat Cantor set. Indeed, its level-n cylinders have total length at most

$$M_n \frac{1}{Q_n} = \frac{(n+1)!}{2^{n-1}(n+2)!} = \frac{1}{2^{n-1}(n+2)} \longrightarrow 0.$$

Thus C has Lebesgue measure zero despite having Hausdorff dimension one. The argument does not determine the Lebesgue measure of the full boundary ∂E and therefore does not claim to resolve the separate positive-measure boundary problem.

8.2 Why the variable radix is essential here

For a fixed parameter ℓ, the known family K_ℓ has boundary dimension $\log(2\ell+1)/\log(2\ell+2) < 1$ [4]. The present construction allows the number of protected branches to increase with the scale. At block n, the protected branch count is $n+1$ while the radix is $2n+4$. The fixed loss factor is asymptotically $1/2$ per level, but its logarithmic cost is only linear in n; the factorial growth of the denominator contributes order $n \log n$. The relative loss therefore vanishes in dimension.

8.3 What has and has not been certified

All infinite assertions in the manuscript are proved symbolically. The supplementary script verifies, with exact integer and rational arithmetic, the complete-residue property for initial blocks, the mixed-radix bijection through level four, monotonicity across initial blocks, and finite portions of the tail inequalities. These computations are consistency checks, not a replacement for proof.

The result has not been checked by an independent mathematician. The most important points for external review are:

1. the use of the complete-residue bijection in the Baire-category interior argument;
2. the isolation lemma and the global validity of the local gaps J_n;
3. the cylinder-count estimates in the mass-distribution proof;
4. the exact applicability of the Guthrie–Nymann classification theorem;
5. possible prior art equivalent to the variable-base construction.

Institutional disclaimer and computational research declaration

This manuscript is an exploratory research preprint issued by the Metriq PRISM Laboratory, an interdisciplinary research laboratory of Metriq Foundation, Inc. PRISM—the Program for Research in Intelligent Systems and Methods—conducts research involving computational, mathematical, scientific, and machine-assisted methods.

The work was developed as part of PRISM's experimental research program in computationally assisted mathematics. Metriq Foundation, acting through the PRISM Laboratory, directed the research process, evaluated the resulting argument, determined the form and scope of the manuscript, and accepts responsibility for its publication as a candidate result.

Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

Data, code, and reproducibility

No empirical dataset was used. The source package accompanying this preprint contains the LuaLaTeX manuscript, exact-arithmetic verification script, figure-generation script, generated figures, and a recorded verification output. The cover artwork is derived from the actual level-1 through level-7 cylinders of the set C in (9); it is not a generic stock image.

Conflicts of interest

Metriq Foundation, Inc. declares no financial conflict of interest concerning the mathematical result presented in this preprint.

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